

## SCR2043 OPERATING SYSTEMS

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Marks

This lab assessment is designed to test your understanding and skills on some basic concepts and tools related to process monitoring and management in operating system. Please follow the instructions carefully and submit your answers in this word document and rename the file as **os-lab-assessment02-studentname-matricno.docx**.

### Essential Steps Before Starting Lab Assessment 2:

#### 1. Download necessary source codes:

Use the `wget` command to retrieve the following source code files to your Linux (or WSL or MacOS) environment:

```
wget -O mainprocess.c https://rebrand.ly/mainprocess_c
wget -O subprocess1.c https://rebrand.ly/subprocess1_c
wget -O subprocess2.c https://rebrand.ly/subprocess2_c
```

#### 2. Compile the source files:

Use the `gcc` compiler to create executable files from the source code.

```
gcc mainprocess.c -o mainprocess
gcc subprocess1.c -o subprocess1
gcc subprocess2.c -o subprocess2
```

#### 3. Execute the dummy processes:

Run all the dummy processes

```
./mainprocess &
```

Press **enter** two times.

4. The dummy processes are running for 2 hours. If you took longer than 2 hours on questions 1-9, please restart the main process with `./mainprocess &`.

## Lab Assessment 2 : Linux Process Monitoring and Management

### Instructions:

1. Carefully execute each command as instructed in the questions.
2. Write down the exact command used for each task.
3. Capture a screenshot of the command's output.

### Question 1

Use the `ps` command with the appropriate option to display a complete list of all running processes within the Linux operating system.

| Command            |       |          |             |
|--------------------|-------|----------|-------------|
| <code>ps -e</code> |       |          |             |
| Output             |       |          |             |
| 3143               | pts/1 | 00:00:00 | mainprocess |
| 3144               | pts/1 | 00:00:00 | mainprocess |
| 3145               | pts/1 | 00:00:00 | mainprocess |
| 3146               | pts/1 | 00:00:00 | subprocess2 |
| 3147               | pts/1 | 00:00:00 | subprocess1 |
| 3148               | pts/1 | 00:00:00 | subprocess2 |
| 3149               | pts/1 | 00:00:00 | subprocess2 |
| 3150               | pts/1 | 00:00:00 | subprocess1 |
| 3152               | pts/1 | 00:00:00 | ps          |

### Question 2

Employ the `ps` command with necessary options to unveil comprehensive details about each running process.

| Command             |      |      |   |       |       |          |               |  |  |
|---------------------|------|------|---|-------|-------|----------|---------------|--|--|
| <code>ps -ef</code> |      |      |   |       |       |          |               |  |  |
| Output              |      |      |   |       |       |          |               |  |  |
| joanne              | 3143 | 3113 | 0 | 15:21 | pts/1 | 00:00:00 | ./mainprocess |  |  |
| joanne              | 3144 | 3143 | 0 | 15:21 | pts/1 | 00:00:00 | ./mainprocess |  |  |
| joanne              | 3145 | 3143 | 0 | 15:21 | pts/1 | 00:00:00 | ./mainprocess |  |  |
| joanne              | 3146 | 3145 | 0 | 15:21 | pts/1 | 00:00:00 | ./subprocess2 |  |  |
| joanne              | 3147 | 3144 | 0 | 15:21 | pts/1 | 00:00:00 | ./subprocess1 |  |  |
| joanne              | 3148 | 3145 | 0 | 15:21 | pts/1 | 00:00:00 | ./subprocess2 |  |  |
| joanne              | 3149 | 3145 | 0 | 15:21 | pts/1 | 00:00:00 | ./subprocess2 |  |  |
| joanne              | 3150 | 3144 | 0 | 15:21 | pts/1 | 00:00:00 | ./subprocess1 |  |  |
| joanne              | 3153 | 3113 | 0 | 15:22 | pts/1 | 00:00:00 | ps -ef        |  |  |

### Question 3

Use the `ps` command with some tools to only list processes named "subprocess" and show some info about them.

| Command                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <code>ps -ef   grep -E 'subprocess'</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |
| Output                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |
| <pre>joanne@secr2043:~\$ ps -ef   grep -E 'subprocess' joanne      3146      3145  0 15:21 pts/1    00:00:00 ./subprocess2 joanne      3147      3144  0 15:21 pts/1    00:00:00 ./subprocess1 joanne      3148      3145  0 15:21 pts/1    00:00:00 ./subprocess2 joanne      3149      3145  0 15:21 pts/1    00:00:00 ./subprocess2 joanne      3150      3144  0 15:21 pts/1    00:00:00 ./subprocess1 joanne      3200      3113  0 15:27 pts/1    00:00:00 grep --color=auto -E subprocess</pre> |  |

### Question 4

Execute the `ps` command, specifying options that reveal only the following columns:

- Process ID (pid)
- Owner of the process (user)
- CPU percentage (pcpu)
- Memory percentage (pmem)
- Command (cmd)

| Command                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <code>ps -eo pid,user,pcpu,pmem,cmd   grep -E 'mainprocess subprocess'</code>                                                                                                                                                                                                                                                                                                                                                                                                       |  |
| Output                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
| <pre>joanne@secr2043:~\$ ps -eo pid,user,pcpu,pmem,cmd   grep -E 'mainprocess subprocess'  3143 joanne   0.0  0.1 ./mainprocess  3144 joanne   0.0  0.0 ./mainprocess  3145 joanne   0.0  0.0 ./mainprocess  3146 joanne   0.0  0.1 ./subprocess2  3147 joanne   0.0  0.1 ./subprocess1  3148 joanne   0.0  0.1 ./subprocess2  3149 joanne   0.0  0.1 ./subprocess2  3150 joanne   0.0  0.1 ./subprocess1  3218 joanne   0.0  0.1 grep --color=auto -E mainprocess subprocess</pre> |  |

### Question 5

Building on the `ps` command used in Question 4, can you add an option to sort the listed processes by their memory usage (`pmem`)?

| Command                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>ps -eo pid,user,pcpu,pmem,cmd --sort=pmem   grep -E 'subprocess'</code>                                                                                                                                                                                                                                                 |
| Output                                                                                                                                                                                                                                                                                                                        |
| <pre>joanne@secr2043:~\$ ps -eo pid,user,pcpu,pmem,cmd --sort=pmem   grep -E 'subprocess' 3146 joanne 0.0 0.1 ./subprocess2 3147 joanne 0.0 0.1 ./subprocess1 3148 joanne 0.0 0.1 ./subprocess2 3149 joanne 0.0 0.1 ./subprocess2 3150 joanne 0.0 0.1 ./subprocess1 3225 joanne 0.0 0.1 grep --color=auto -E subprocess</pre> |

### Question 6

Construct a command using `ps`, suitable options, and any additional tools to visualize the hierarchical structure (tree-like) of the following processes:

- "mainprocess"
- "subprocess1"
- "subprocess2"

| Command                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>ps --forest -C mainprocess -C subprocess1 -C subprocess2</code>                                                                                                                                                                                                                                                                                                                                                                                                        |
| Output                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <pre>joanne@secr2043:~\$ ps --forest -C mainprocess -C subprocess1 -C subprocess2   PID TTY          TIME CMD  3143 pts/1        00:00:00 mainprocess  3144 pts/1        00:00:00 \_ mainprocess  3147 pts/1        00:00:00   \_ subprocess1  3150 pts/1        00:00:00   \_ subprocess1  3145 pts/1        00:00:00 \_ mainprocess  3146 pts/1        00:00:00 \_ subprocess2  3148 pts/1        00:00:00 \_ subprocess2  3149 pts/1        00:00:00 \_ subprocess2</pre> |

### Question 7

Use `pstree` command with option that show the number of threads to each process.

| Command                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>pstree -T pstree -T -s 3143</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Output                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <pre>joanne@secr2043:~\$ pstree -T systemd+-ModemManager   -2*[agetty]   -cron   -dbus-daemon   -multipathd   -polkitd   -rsyslogd   -sshd+-sshd---sshd---bash           '-sshd---sshd---bash+-mainprocess+-mainprocess---2*[subprocess1]                                                                       '-mainprocess---3*[subprocess2]                                   '-pstree   -systemd-journal   -systemd-logind   -systemd-network   -systemd-resolve   -systemd-timesyn   -systemd-udev   -udisksd   -unattended-upgr</pre> |
| <pre>joanne@secr2043:~\$ pstree -T -s 3143 systemd---sshd---sshd---sshd---bash---mainprocess+-mainprocess---2*[subprocess1]     '-mainprocess---3*[subprocess2]</pre>                                                                                                                                                                                                                                                                                                                                                                        |

### Question 8

Use `renice` command to change priority level of one of process “subprocess1”.

| Command                                                                                                                            |
|------------------------------------------------------------------------------------------------------------------------------------|
| <pre>sudo renice -10 3143</pre>                                                                                                    |
| Output                                                                                                                             |
| <pre>joanne@secr2043:~\$ sudo renice -10 3143 [sudo] password for joanne: 3143 (process ID) old priority 0, new priority -10</pre> |

### Question 9

Terminate all running processes with the name “mainprocess”.

| Command                                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| killall mainprocess                                                                                                                                                                                                                                                               |
| Output                                                                                                                                                                                                                                                                            |
| <pre>joanne@secur2043:~\$ killall mainprocess Main process (ID: 1011) received signal: 15. Terminating... Main process (ID: 1010) received signal: 15. Terminating... Main process (ID: 1012) received signal: 15. Terminating... [1]+  Done                  ./mainprocess</pre> |

### Question 10

Write a short C or Python code (choose only one language) demonstrating multiprocessing with `fork()` and `wait()`. Compile and/or run the code. Show the output.

Source Code:

```
GNU nano 7.2 myProgram.c *
#include<stdio.h>
#include<unistd.h>
#include<sys/wait.h>

int main(){
    pid_t pid=fork();
    if (pid == -1){
        perror("Oops ,fail");
        return 1;
    }
    if(pid == 0){
        printf("Child Process:PID = %d, Parent PID",getpid(),getppid());
        _exit(0);
    }else{
        printf("Parent Process: PID = %d ,Child PID = %d\n",getpid(),pid);
        wait(NULL);
        printf("Parent Porcess: Child has terminated\n");
    }
    return 0;
}
```

Output:

```
joanne@secr2043:~$ gcc -o myProgram myProgram.c
joanne@secr2043:~$ ./myProgram
Child Process:PID = 2702, Parent PID = 2701
Parent Process: PID = 2701 ,Child PID = 2702
Parent Porcess: Child has terminated
joanne@secr2043:~$
```